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Automated Technology for Tubular Connection Integrity Assurance: Enhancing Safety, Well Integrity, Operational Efficiency, and Reducing Co_2 Emissions on Artificial Island

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Abstract

This study evaluates the efficiencies of an automated connection integrity system compared to traditional non-automated equipment under identical work scopes and operating conditions. The comparison focuses on four key performance indicators (KPIs): Safety, Quality, Operational Efficiency, and Environmental Impact. To ensure data quality, trials were conducted over four wells, running both lower and upper completions on the same rig to maintain operational consistency and facilitate technology adoption by the crews.

Historical data from running lower and upper completion strings on the selected rig was analyzed to derive specific benchmark KPIs, including Red Zone Exposure with hands-free positioning, Personnel on Board (POB), Final Torque standard deviation, Damaged Threads, Connection time, rig up and rig down time, and CO2 emissions. Traditional methods and equipment were replaced with automated technology, and the same KPIs were monitored for four consecutive wells on the same rig.

Data analysis indicates significant improvements in red zone management, with personnel exposure during connection makeup reduced by 80%. There was an absolute reduction in damaged threads, achieving zero damaged threads. Connection time, measured in real-time by the rig, improved by an average of 12%. Rig up and rig down times were reduced by an average of 15%, and CO2 emissions decreased by 12%. The automated connection integrity system, combined with the hands-free positioning device, significantly enhances safety, quality, operational efficiency, and environmental impact by providing consistent execution with minimal deviations.

Introduction

The oil and gas industry continues to pursue technological innovations that enhance operational safety, improve integrity, and reduce environmental impact while maintaining cost-effectiveness. Traditional tubular connection makeup processes rely heavily on human intervention, creating inherent variability in connection quality and exposing personnel to hazardous operational zones. This dependency on manual operations introduces inconsistencies that can compromise well integrity and operational efficiency.

The concept of automated connection integrity systems represents a paradigm shift from conventional practices, much like the evolution from manual drilling to automated drilling systems. By eliminating human variability, automated connection makeup systems promise to revolutionize tubular running operations by providing consistent, repeatable performance independent of human factors.

This paper presents a comprehensive evaluation of an integrated automated connection integrity system comprising automated makeup control algorithms, autonomous connection evaluation software, and hands-free positioning technology. The system was evaluated against traditional equipment under identical operational conditions on artificial island drilling operations, providing a controlled environment for performance comparison.

The data was taken from rig real time connection logs, to ensure same data points are used to compare the traditional method with automated equipment.

History

Traditional tubular connection makeup has been the industry standard for decades, relying on operator experience for manual control of power tong to apply torque and hydraulic dump valves to stop the rotation at target torque. This method was fully relying on the operator to control the speed of the power tong manually which was very inconsistent with a risk exceeding the OCTG OEM speed recommendation and damaging the thread.

The hydraulic dump valve with a sudden hydraulic power cut off could still result with over torque due to inertia and the risk is higher with higher speed. The connection evaluation, which is the most critical part of well integrity, was completely reliant on human judgement and decision making, where one unintentional oversight can lead to catastrophic consequences up to production loss.

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Increased personnel exposure in hazardous zones, requiring up to 8 people in red zone during connection make up is one of the most intensive red zone operations on the rig floor, which is not only a safety hazard, but also a very labor-intensive operation that occupies work force to manually position the power tong on well center for each connection.

The environmental impact of less efficient traditional methods which will prolong rig time and increase fuel consumption was often overlooked, with additional impact of damaged pipes and personnel transportation.

Recent developments in connection integrity technology have focused on three primary areas: automated torque control, real-time connection evaluation, and remote positioning systems. The integration of these technologies represents a significant advancement in tubular running operations, addressing long-standing industry challenges related to safety, quality, efficiency and environmental impact.

Figure 1—Personnel Red Zone exposure in traditional operations

Automated Connection Integrity Technology

Automated Connection Makeup System

The automated connection makeup system utilizes advanced speed-control algorithms that employ continuous torque measurements to automatically regulate makeup velocity as the connection approaches optimum torque. This technology eliminates the need for traditional dump valves and associated sudden stops in the makeup process, providing smooth, controlled connection makeup.

The system employs dual displacement motors and hydraulic valve sections that enable precise control of connection makeup speed. Real-time torque monitoring and adaptive speed control regulate both torque application and revolutions per minute throughout the connection process. This precise control technology ensures consistent connection makeup independent of operator-specific influences, with a smooth stop at target torque, preventing any under torque or over torque.

Autonomous Connection Evaluation

The autonomous evaluation subsystem analyzes connection data with resolution 10 times more refined than traditional visual interpretation methods. The system automatically detects torque-shoulder positions, eliminating manual data adjustments typically required by operators. Integrated evaluation software assesses connections according to OCTG OEM criteria, providing objective pass/fail determinations. The evaluation process takes 5 seconds with absolute certainty, processing multiple parameters simultaneously to verify connection integrity.

Hands-Free Positioning Technology

The positioning system incorporates a telescopic mechanism with remote operation capabilities. The system features a rigid clamp for well center adjustment and retracts to parking position between operations. This configuration provides operator-friendly operation with minimal footprint requirements while maintaining fast running speeds due to precision and repeatability, enabling one man operation instead of up to eight people in red zone.

Evaluation Methodology

Experimental Design

The evaluation was conducted using a controlled comparison methodology on artificial island operations. Four consecutive wells were selected for the trial, with both lower and upper completion strings run using the same drilling rig to maintain operational consistency and same well design to maintain operational requirements, changing only one variable – traditional equipment with automated equipment. This approach eliminated variables associated with different rig configurations, crew capabilities, and operational procedures.

Historical data from traditional connection operations on the same rig provided baseline performance metrics. The automated system was then deployed for four consecutive wells, with identical KPIs monitored and recorded. This methodology ensured direct comparability between traditional and automated approaches under equivalent operational conditions.

All the data points were taken from rig sensors to ensure consistency and avoid subjective interpretation, automating the results outputs.

Data Collection and Analysis

Performance data was collected across four primary categories as indicated in Table 1.

Table 1—Key Performance Indicators

Category	KPI (Average per Well)	Traditional	AutomationTarget	AutomationActual	Improvement
Safety	Red Zone Exposure Reduction [Man Hours]	211	42	42	80%
	POB Reduction	5	4	4	20%
	Hands-Free Power Tong Positioning	No	Yes	Yes	100%
Quality	Automated Connection Make Up	No	Yes	Yes	100%
	Automated Connection Evaluation	No	Yes	Yes	100%
	Final Torque Standard Deviation [ft-lb]	227	34	21	91%
	Damaged Threads Reduction	1.26%	0.63%	0	100%
Operational Efficiency	Connection Time [min]	7.5	6.75	6.6	12%
	Laid Down Joints	2.5	1.25	0	100%
	Rig Up Time [Hours]	1.66	1.49	1.41	15%
	Rig Down Time [Hours]	1.2	1.08	1.03	15%
Environment	CO_2 Emissions Reduction [kg]	53,510	48,159	47,089	12%

Data collection utilized rig-mounted real-time monitoring systems, automated data logging, and standardized reporting protocols. Statistical analysis was performed to ensure data validity and significance of observed improvements.

Results and Discussion

Safety Performance. The automated system demonstrated substantial safety improvements across all measured parameters. Red zone exposure during connection makeup was reduced by 80%, from 211 man-hours (traditional) to 42 man-hours (automated) per well. This reduction was achieved through the hands-free positioning system, which eliminated the need for personnel to manually guide tubulars during connection makeup. The hands-free positioning capability was particularly beneficial during high-temperature operating conditions, reducing heat-related stress on crew members while maintaining operational efficiency.

Personnel on board (POB) requirements decreased by 20%, from an average of 5 to consistent 4 TRS personnel per operation, 2 per 12 hours shift.

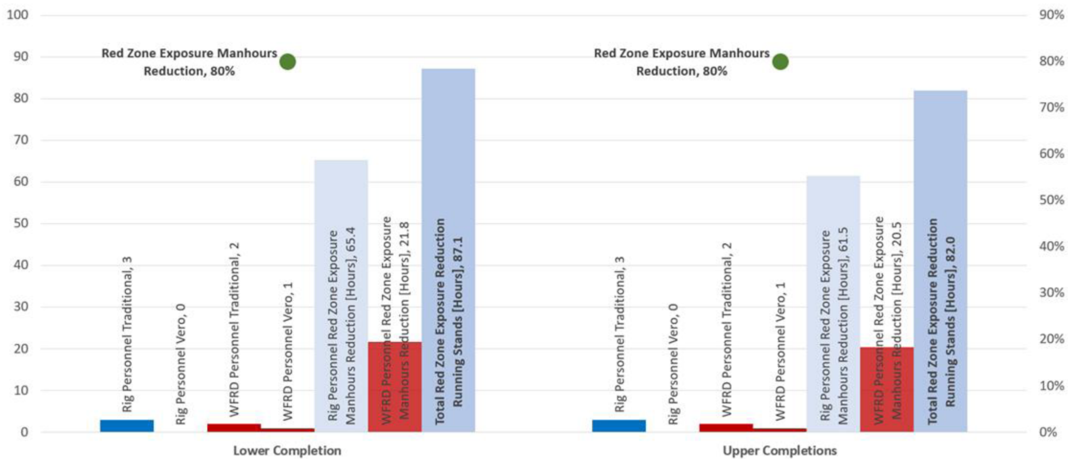

Figure 2—Personnel Red Zone exposure Reduction

The safety improvements are not only removing human operators from hazardous zones and dramatically reducing incident potential but also improved operational efficiency with precise power tong positioning.

Tubular Connection Quality. Connection quality metrics showed remarkable improvements under automated control. Final torque standard deviation decreased by 91%, from 227 ft-lb (traditional) to 21 ft-lb (automated). This improvement demonstrates the superior consistency achievable through automated torque control compared to manual operations which indicates a reduction in damaged or over/under torqued connections and gives assurance of connection integrity.

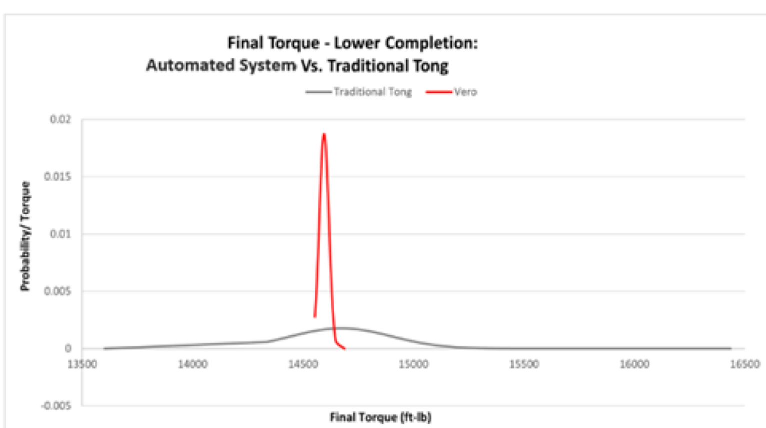

Figure 3—Final Torque Standard Deviation

Damaged thread occurrences were eliminated, achieving zero damaged connections across all automated operations compared to a 1.26% damage rate with traditional methods. This improvement not only enhances well integrity but also eliminates costly connection remediation operations which includes damaged tubulars transportation, storage and threads re-cut or scraping the tubulars.

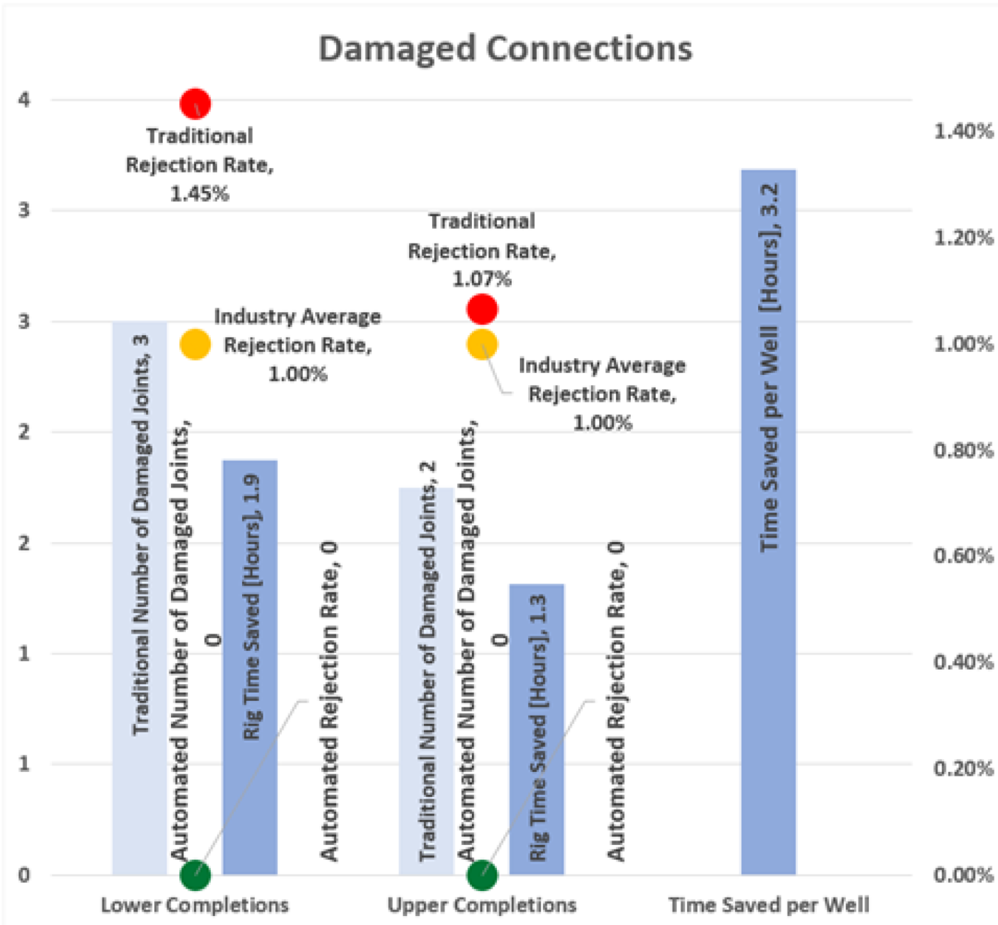

Figure 4—Damaged Connection Comparison Chart

The connection quality improvements clearly demonstrate how computer-controlled systems consistently outperform manual operations in terms of accuracy and repeatability, achieving overall operational efficiency improvements.

Operational Efficiency. Connection time improved by an average of 12%, decreasing from 7.5 minutes (traditional) to 6.6 minutes (automated) per connection. This improvement was achieved through optimized speed control algorithms that eliminate connection re-runs which are often occurring in manual operations while maintaining connection integrity.

Rig up and rig down times were reduced by 15%, from 1.66 hours to 1.41 hours and 1.2 hours to 1.03 hours, respectively. These improvements result from streamlined equipment positioning to rig floor before actual TRS operations and reduced equipment requirements by eliminating the traditional JAM system and dump valve.

The elimination of laid down joints, from 2.5 traditional to 0 automated, further enhanced operational efficiency by reducing non-productive time associated with connection remediation.

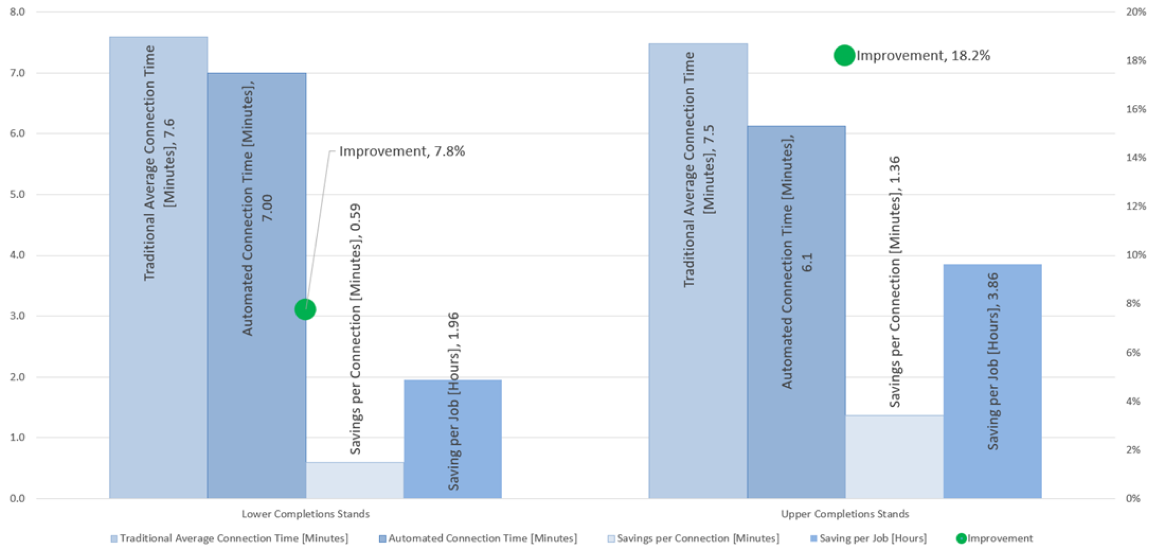

Figure 5—Connection time comparison chart

Environmental Impact. CO_2 emissions were reduced by 12%, from 53,510 kg (traditional) to 47,089 kg (automated) per well. This reduction results from improved operational efficiency, reduced equipment operating time, and elimination of remedial operations which resulted with reduced rig time and rig consequently rig fuel consumption, not including savings from personnel transportation and damaged tubulars transportation for simplicity of this study.

The environmental benefits extend beyond direct emissions reduction, as improved connection integrity reduces the risk of wellbore integrity issues that could lead to environmental incidents and additional CO_2 emissions.

Learning Curve Analysis. The implementation demonstrated a typical technology adoption learning curve, with performance improvements accelerating after initial system optimization. Key learning elements included:

- Adjustment of system mounting configurations
- Optimization of standby positioning for efficient movement
- Valve system modifications for enhanced positioning speed
- Crew synchronization and procedural standardization

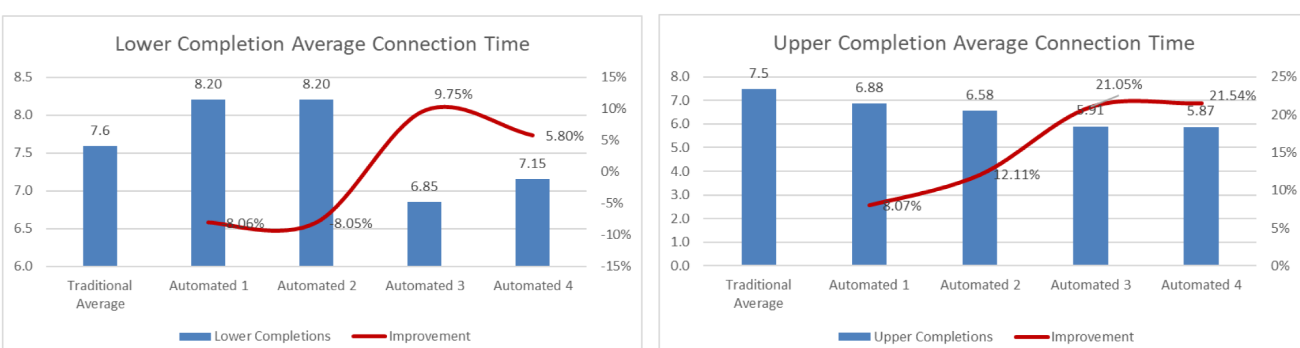

Figure 6—Lower and Upper Completions Learning Curves

Continuous performance improvement was observed across all four wells, with the most significant gains achieved in the final two wells after crew familiarity and system optimization were completed.

Economic Impact of Automated Connection Integrity

The economic analysis demonstrates substantial value creation through automated connection integrity implementation. For artificial island operations, the system provided an average savings of 8.7 hours per well.

For jack-up rig operations, savings increase to 15.73 hours per well. The enhanced value for jack-up operations results from running tubulars in singles which increases numbers of connections made on rig floor, amplifying the impact of efficiency improvements.

The technology demonstrates compelling return on investment characteristics:

- Tangible: Time savings per well are higher than additional cost for automated equipment,
- Risk mitigation: Enhanced integrity through consistent connection quality preventing costly interventions.

The results of this study are confirming that automation can deliver safety enhancements, long term well integrity and operational efficiency which makes it a sustainable investment with long term outlook.

Novel Technical Contributions

The primary technical novelty lies in the seamless integration of three distinct automation technologies:

1. Automated Makeup Control: Eliminates human influence in final torque application through continuous monitoring and adaptive speed control
2. Autonomous Connection Evaluation: Provides objective connection assessment based on OEM criteria without subjective interpretation
3. Hands-Free Positioning: Removes personnel from hazardous zones while maintaining operational efficiency

The speed-control algorithm represents a significant advancement over traditional dump valve systems. By utilizing continuous torque feedback to modulate makeup speed, the system eliminates sudden stops and provides smooth torque application throughout the connection process.

The autonomous evaluation system processes connection data with unprecedented resolution, providing immediate feedback on connection quality. This capability enables real-time quality assurance rather than post-operation analysis, allowing for immediate corrective action if required.

Enhanced connection integrity reduces the risk of wellbore integrity failures that could result in environmental incidents. The consistent connection quality provided by automated systems serves as a preventive measure against potential hydrocarbon releases.

Future Development Opportunities

The automated connection integrity platform provides a foundation for further technological advancement for fully Remote Operations and completely removing personnel from red zone, which can be achieved with Wi-Fi controlled systems. This would enable personnel to operate tubular running equipment from the driller's cabin for further safety and efficiency enhancements. A more advanced option would be full integration into the driller's chair which would enable rig contractors to run tubular equipment with remote technical support from equipment manufacturers.

Conclusions

The comprehensive evaluation of automated connection integrity technology demonstrates substantial improvements across all measured performance indicators. The integration of automated makeup control, autonomous evaluation algorithms, and hands-free positioning creates a synergistic effect that enhances safety, quality, operational efficiency, and environmental performance.

Automated Connection Integrity Technology demonstrates scalability across various operational environments:

- Offshore drilling operations
- Land drilling operations
- Deep Water drilling operations

The automated connection integrity system represents a significant technological advancement that addresses fundamental challenges in tubular running operations. The simplicity of the system makes the learning curve very efficient and reduces the reliance on highly skilled and experienced personnel which opens opportunities for rig contractors to run tubulars.

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